Beyond Infinity: Two-Stage Sounding Rocket for Mustafar

OpenRocket report | Files: BeyondInfinity_Rocket.ork, BeyondInfinity_FlightData.csv | Body material: **cardboard**

1. Compliance with the problem statement (Run_Mustafar)

Requirement	Achieved	Status
Apogee 3900 to 4100 m	**4038 m**	SATISFIED
Payload 0.95 to 1.05 kg	**1.000 kg**	SATISFIED
Length <= 2200 mm	**1550 mm**, max diameter 8 cm	SATISFIED
Total mass <= 10 kg (cardboard)	**3.218 kg** with motors (2.049 kg without)	SATISFIED
Total impulse <= 2800 N·s	**994 N·s** (476 + 518)	SATISFIED
Drogue at apogee	Opens at 4038 m, t = 32 s	SATISFIED
Main chute deployment	Opens at 2000 m on descent, t = 197 s (as clarified; PS table says apogee)	AS CLARIFIED
Payload deployment at apogee	Payload stays inside the sustainer. OpenRocket cannot release it from a stage, so it comes down on the sustainer chutes	MODELLED
Parachute recovery	Booster chute, drogue and main chute, all stages recovered	SATISFIED
Descent velocity 4 to 8 m/s	Main chute phase: sustainer **5.5 m/s**, booster **5.3 m/s** (vertical)	SATISFIED
Mustafar environment	6.50 m/s², 75 kPa, 700 K, wind 10 m/s at 270°, 0 m site. Density follows from P and T (0.373 kg/m³)	APPLIED
No manual mass overrides	All masses come from OpenRocket material and motor definitions	SATISFIED

Stability (no PS limit, reported as simulated): **4.32 cal (22.3 %)**, CG 76.7 cm, CP 111 cm, M = 0.3, full stack at lift-off.

2. Vehicle and design process

Part	Specification	Material
Nose cone	Ogive, 15 cm long, 8 cm base, wall 0.2 cm	Polystyrene
Sustainer body	80 cm long, OD 8 cm, ID 7.6 cm, wall 0.2 cm	Cardboard
Payload	1000 g mass component, 7 cm x 15 cm, top of sustainer body	n/a
Sustainer fins	4 trapezoidal: root 10, tip 5, height 6, sweep 5, thick 0.3 cm	Birch plywood
Sustainer motor	518I165-17A (518 N·s), mount 30 cm, plugged, lights 1 s after booster burnout	Cardboard mount
Booster body	60 cm long, OD 8 cm, wall 0.2 cm	Cardboard
Booster fins	4 trapezoidal: root 15, tip 7, height 8, sweep 8, thick 0.3 cm	Birch plywood
Booster motor	475I445-16A (476 N·s, 1.07 s burn), mount 35 cm, plugged, lights at launch	Cardboard mount
Parachutes	Main ____ cm at 2000 m. Drogue 80 cm at apogee. Booster 120 cm at booster apogee	Ripstop nylon

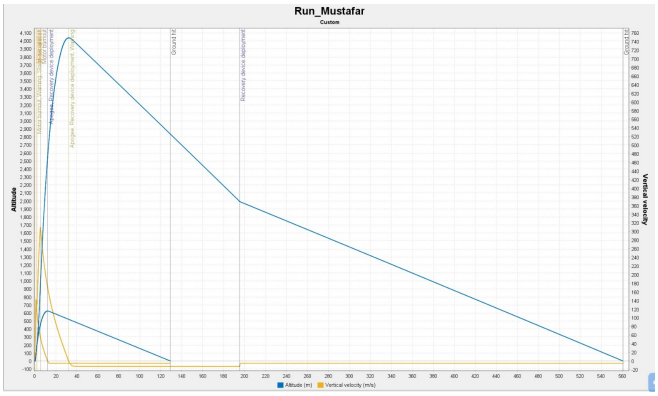
- **Size:** the first model had a 5 cm tube and a 2.5 cm block holding 1 kg. We moved to an 8 cm tube and a 7 x 15 cm payload block so the payload and 54 mm motor mounts fit.
- **Staging:** separation was first tied to the motor ejection charge, which our plugged motors never fire. It now happens at booster burnout.
- **Wind:** at 10.9 m/s off the rod the 10 m/s wind tipped the rocket over (3599 m, 52 m/s sideways at deployment). Without wind the same design reached 3998 m. We then tuned the motors until the Mustafar run gave 4038 m.
- **Recovery:** drogue 50 to 80 cm, booster chute 140 to 120 cm, main chute moved from apogee to 2000 m. Dry mass changed from 2056 to 2049 g and apogee from 4066 to 4038 m. Wind turbulence is 0 in the comparison runs so they can be repeated.

3. Recovery and descent velocity

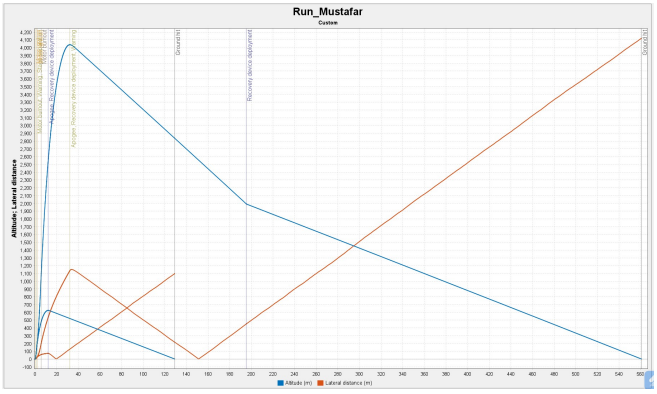
The booster chute opens at the booster apogee (625 m, t = 12 s). The drogue opens at the sustainer apogee. The main chute opens at 2000 m. Both motors are plugged, so a real build needs a timer or flight computer to fire the chutes.

Phase (from Plot 1)	Calculation	Vertical speed
Sustainer under main chute: 2000 m at t = 197 s to ground at t = 561 s	2000 m / 364 s	**5.5 m/s**
Booster under chute: 625 m at t = 12 s to ground at t = 129 s	625 m / 117 s	**5.3 m/s**
Sustainer under drogue only: 4038 m to 2000 m, t = 32 s to 197 s	2040 m / 165 s	12.4 m/s

The PS does not say whether 4 to 8 m/s is vertical or total speed. We use vertical speed, since a 10 m/s wind keeps total speed above 10 m/s. The 10.9 m/s ground hit value in OpenRocket is total speed: sqrt(10.9² - 5.5²) = 9.4 m/s of it is wind drift. The sustainer lands about 4.1 km from the pad (Plot 2). The drogue phase is faster (12.4 m/s) because the main chute stays closed until 2000 m. The drogue warning at apogee (21.8 m/s) is sideways speed from the wind, since vertical speed is near zero there. The booster note "open forward airframe" only says the booster has no nose cone after separation.



Plot 1: altitude and vertical velocity vs. time



Plot 2: altitude and lateral distance vs. time

4. Earth vs. Mustafar study

Run_Earth is the baseline (15 °C, 1013.25 mbar, no wind, g = 9.79 m/s² from OpenRocket at the launch latitude). Each next run changes one setting. Run_Mustafar changes all of them. Mach is max velocity / sqrt(1.4 R T), calculated. Predicted descent uses v proportional to sqrt(g / density).

Run	Change from Earth	Density kg/m³	Apogee m	Max vel m/s	Mach (calc.)	Time to apogee s	Descent m/s (predicted)	Drift
Run_Earth	baseline	1.225	2007	242	0.71	18.9	3.75	0
Run_Gravity	g = 6.50	1.225	2341	250	0.73	23.5	3.05 (3.06)	0
Run_Temp700K	T = 700 K	0.504	3012	293	0.55	23.8	5.84 (5.85)	0
Run_Pressure75k	P = 75 kPa	0.907	2344	261	0.77	20.7	4.36 (4.36)	0
Run_Wind10	wind 10 m/s, 270°	1.225	1867	240	0.71	18.2	9.2, drogue only	about 2 km (calc.)
Run_Mustafar	all four	0.373	**4038**	317	0.60	32.1	5.5 (5.54)	4.1 km

Variable	Hypothesis	Result	Held?
Gravity 6.50	Less weight, longer and higher climb, slower chute descent (v ~ sqrt g)	+334 m apogee, +4.6 s to apogee, descent 3.75 to 3.05 m/s (ratio 0.81, predicted 0.81)	**Yes**
Temperature 700 K	Thinner air (1.225 to 0.504), less drag, faster descent. Speed of sound sqrt(gamma R T) goes 340 to 530 m/s, so Mach drops	+1005 m apogee, Mach 0.71 to 0.55, descent 3.75 to 5.84 m/s (predicted 5.85)	**Yes**
Pressure 75 kPa	Density falls 26 %, less drag, higher apogee, faster descent	+337 m apogee, descent 3.75 to 4.36 m/s (predicted 4.36)	**Yes**
Density 0.374	OpenRocket takes density from P and T, so it is tested through them. Effects should compound	Single gains sum to +1676 m. All four together give +2031 m. Descent 5.5 m/s (predicted 5.54)	**Yes**
Wind 10 m/s	Weathercocking loses altitude, drift near wind speed, higher speed at deployment	-140 m apogee, deployment speed 9.67 to 23.7 m/s, 9.6 m/s horizontal at landing. Apogee under 2000 m, so the main chute never opened	**Yes**

Limits: the 6.50 m/s² gravity was applied with the planetary environment extension, so it must be installed to repeat the Mustafar and gravity runs. OpenRocket shows Mach 0.565 at 317 m/s in Run_Mustafar (speed of sound about 561 m/s), against 530 m/s calculated for air at 700 K. We have not explained the 6 % gap. The CSV holds the sustainer only. Wind direction 270° was entered as OpenRocket's compass direction.